ARUSHI Y
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PROFESSIONAL SUMMARY:

Data Engineer specializing in AI powered data engineering, PySpark, Databricks, Delta Lake, and production lakehouse architectures, with experience developing scalable Bronze,Silver and Gold pipelines for structured and unstructured enterprise data. Skilled in NLP and LLM assisted information extraction, document understanding, semantic embeddings, entity resolution, record linkage, and canonical data modeling. Experienced in designing provenance-aware data pipelines with source traceability, confidence-based validation, trust classifications, data quality gates, and quarantine workflows. Strong background in Apache Airflow orchestration, freshness SLA management, distributed Spark optimization, data governance, Unity Catalog, and high performance analytical serving layers. Experienced in integrating GenAI capabilities, including RAG, Databricks Mosaic AI, and Vector Search, into governed enterprise data platforms.

EDUCATION:

University of North Texas — Denton, TX
Master of Science in Information Systems and Technologies – **GPA 3.8/4.0** **Aug 2022–May 2024**
Sri Venkateswara University – Tirupati, Andhra Pradesh
Bachelors of Technology in Electronics and Communication – **GPA 3.2/4.0** **Aug 2015 – May 2019**

SKILLS:

Programming & Data Processing: Python, PySpark, SQL, Scala, Shell Scripting, Apache Spark, Spark SQL, Pandas, NumPy.
Cloud & Lakehouse Platforms: Azure Databricks, Azure Data Lake Storage Gen2, Delta Lake, Microsoft Fabric, OneLake, Azure Synapse Analytics, Amazon S3, AWS Glue, Amazon EMR, Snowflake, Redshift, BigQuery.
Medallion Architecture & Data Engineering: Bronze/Silver/Gold Architecture, Lakehouse Design, Distributed Processing, Incremental Data Processing, Delta Tables, Schema Evolution, CDC, Data Standardization, Data Modeling, ETL/ELT, Data Contracts.
AI, NLP & Document Intelligence: Natural Language Processing, Large Language Models, LLM-Assisted Extraction, Document Understanding, HTML Parsing, Structured Attribute Extraction, Semantic Embeddings, Vector Search, Retrieval-Augmented Generation, Databricks Mosaic AI, Snowflake Cortex AI, OpenAI API, Prompt Engineering, Confidence Scoring.
Entity Resolution & Master Data: Entity Resolution, Record Linkage, Canonical Entity Modeling, Deterministic Matching, Probabilistic Matching, Fuzzy Matching, Attribute Normalization, Deduplication, Match Confidence, Survivorship Rules, Cross-Source Reconciliation.
Governance, Provenance & Data Quality: Unity Catalog, Data Lineage, Data Provenance, Trust Tiers, Evidence Tracking, Source-to-Target Mapping, Data Quality Gates, Schema Validation, Exception Handling, Quarantine Pipelines, Audit Metadata, Data Freshness SLAs, RBAC.
Orchestration & Streaming: Apache Airflow, Azure Data Factory, Apache Kafka, Apache Flink, Apache NiFi, AWS Kinesis, AWS Glue, Spark Structured Streaming, Backfills, DAG Management, Workflow Scheduling.
Databases & Analytics: Snowflake, PostgreSQL, MongoDB, MySQL, Oracle, Teradata, Redis, HBase, dbt, Power BI, Tableau, Streamlit.
Spark Performance Optimization: Adaptive Query Execution, Partition Pruning, Broadcast Joins, Shuffle Optimization, Data Skew Mitigation, Delta Lake Compaction, Query Optimization, Executor Memory Tuning, Cluster Autoscaling.
DevOps & Infrastructure: Git, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD, AWS CloudFormation, Ansible.

PROFESSIONAL EXPERIENCE:

Abbott – Alameda, California **Jun 2025 – Present**
Senior Software Engineer

- Architected and optimized production-grade PySpark and Azure Databricks pipelines processing 8M+ records daily, implementing Bronze,Silver, and Gold medallion architecture on Delta Lake and ADLS Gen2 to integrate structured enterprise systems, operational datasets, and unstructured content into governed, analytics-ready Gold data products.
- Developed scalable NLP and LLM assisted information extraction pipelines using Databricks Mosaic AI, Python, document understanding, HTML parsing, semantic embeddings, and prompt-based extraction to transform unstructured documents, web content, and enterprise text into structured business attributes with confidence scoring and evidence-level traceability.
- Engineered high-volume entity resolution and record linkage workflows using PySpark, implementing deterministic and probabilistic matching, fuzzy string similarity, standardized business identifiers, and cross-source attribute reconciliation to consolidate fragmented records into consistent canonical enterprise identities.
- Designed and implemented canonical identity resolution gates with configurable match-confidence thresholds, deduplication logic, survivorship rules, source-priority policies, and exception handling to distinguish verified

identities from ambiguous or unresolved records before promotion into trusted Gold datasets.
- Built source-conformance and standardization frameworks for heterogeneous enterprise data, including CPQ/PIM-style product and operational attributes, applying schema normalization, source-to-target mapping, data contracts, attribute harmonization, and validation rules across structured and LLM-extracted datasets.
- Established record-level data provenance and trust classification controls, capturing original source identifiers, document references, extraction methods, model versions, processing timestamps, transformation lineage, and trust tiers while maintaining explicit separation between extracted evidence and authoritative business facts.
- Implemented automated data quality gates and quarantine frameworks to isolate schema violations, failed entity matches, conflicting attributes, low-confidence LLM extractions, and incomplete records into auditable Delta Lake exception tables with rejection reason codes, remediation workflows, and controlled reprocessing.
- Orchestrated end-to-end ingestion, extraction, entity resolution, validation, and Bronze to Gold transformation workflows using Apache Airflow and Azure Data Factory, implementing DAG dependency management, incremental processing, retries, backfills, automated alerting, freshness SLA monitoring, and failure recovery.
- Optimized Spark and Delta Lake performance through adaptive query execution, partition pruning, broadcast joins, shuffle tuning, executor memory optimization, file compaction, checkpointing, and incremental transformations to support high-throughput processing while preserving reliable data freshness and pipeline scalability.
- Designed high-volume lakehouse storage and lean analytical serving architectures, retaining raw records, historical versions, source-level evidence, and detailed lineage in Delta Lake while exposing compact, trusted, canonical Gold datasets through Databricks SQL, Snowflake, and Power BI for efficient downstream consumption.
- Built governed GenAI-ready lakehouse datasets integrating Databricks Mosaic AI, Retrieval-Augmented Generation (RAG), Vector Search, semantic embeddings, Unity Catalog, and Delta Lake, collaborating with engineering and business stakeholders to establish trusted data definitions, metadata governance, validation requirements, and production-ready AI consumption patterns.

Charter Communications – St.Louis, MO **Aug 2024 – March 2025**
Specialized Software Engineer

- Designed and developed scalable PySpark and Azure Databricks data pipelines using Delta Lake and ADLS Gen2, implementing Bronze, Silver, and Gold processing patterns to ingest, standardize, and curate heterogeneous structured, semi-structured, and unstructured enterprise datasets.
- Built NLP and LLM-assisted document extraction workflows using Python, text preprocessing, HTML parsing, semantic embeddings, and Vector Search to extract structured attributes from enterprise content while retaining source references, extraction metadata, and confidence-based validation information.
- Implemented source provenance, lineage tracking, and trust-aware validation controls capturing ingestion timestamps, source-system identifiers, record references, transformation history, and data quality classifications to support traceability throughout distributed processing workflows.
- Built automated data quality and exception handling processes using PySpark and Delta Lake to identify schema inconsistencies, duplicate records, unresolved mappings, invalid attributes, and incomplete transformations, routing failed records into reviewable quarantine datasets.
- Engineered high-throughput batch and streaming ingestion using Apache Kafka, AWS Kinesis, Azure Data Factory, and Databricks, supporting incremental data movement, late-arriving records, replayability, checkpoint recovery, and reliable downstream data availability.
- Optimized distributed Spark and Delta Lake workloads through adaptive partitioning, shuffle optimization, broadcast joins, data-skew mitigation, executor memory configuration, Delta file compaction, and cluster autoscaling to improve processing stability and query efficiency.
- Built reusable Snowflake and dbt transformation layers, publishing standardized, conformed analytical datasets with consistent business definitions while separating high-volume historical lakehouse storage from optimized serving and reporting structures.
- Developed AI-ready enterprise datasets by integrating structured and unstructured content with embedding generation, semantic retrieval, Snowflake Cortex AI, and Streamlit applications, enabling governed search, AI-assisted querying, and downstream analytical consumption.
- Supported production data orchestration and operational reliability through Azure Data Factory, Databricks workflows, automated validation, dependency management, retry strategies, monitoring, and recovery procedures, ensuring data pipeline stability and timely delivery of curated datasets.

Centene Corporation – Bangalore, India **Apr 2020 – Aug 2022**
Software Engineer

- Architected high-throughput streaming data ingestion platforms using Kafka, Apache NiFi, and HBase, implementing durable processing, late-data handling, watermarking, fault tolerance, and replay mechanisms for enterprise telemetry workloads.
- Built scalable AWS data engineering pipelines using S3, EMR, Glue, Lambda, Kinesis, and Redshift, enabling distributed processing, incremental ingestion, automated transformation, and downstream analytical consumption.
- Integrated Snowflake, BigQuery, Redis, and enterprise analytical environments, implementing reusable

transformation patterns, governed access controls, audit ready metadata, and optimized data-access strategies for efficient reporting and downstream business consumption.

- Automated AWS data pipeline orchestration using AWS Step Functions, EventBridge, and CloudWatch, coordinating scheduled ETL workflows, configuring event-driven triggers, and implementing centralized logging, monitoring, and failure notifications to improve pipeline reliability and operational efficiency.

Wells Fargo – Bangalore,India **Jun 2019 – Mar 2020**
Software Engineer

- Contributed to enterprise data platform foundations by onboarding source systems, mapping schemas, and publishing curated datasets into shared warehouse layers.

- Developed scalable GCP based ETL pipelines using Python, PySpark, Google Cloud Storage (GCS), and BigQuery**,** integrating enterprise data sources and transforming raw datasets into structured analytical tables.
- Implemented BigQuery data warehouse solutions using partitioned tables, clustered storage, optimized SQL queries, and incremental loading strategies to improve query performance and processing efficiency.
- Built and maintained Apache Airflow DAGs to orchestrate scheduled batch ingestion and transformation workflows, managing task dependencies, automated retries, backfills, and pipeline failure recovery.
- Developed PySpark based batch processing workflows on Google Cloud Dataproc, performing distributed data transformations and preparing large datasets for downstream BigQuery analytics.
- Implemented GCP data security and access controls using Cloud IAM, managing service account permissions and role-based access to Cloud Storage and BigQuery datasets to support secure enterprise data processing.

Wells Fargo – Bangalore,India **Feb 2018 – Jun 2019**
Software Engineer Intern

- Assisted in developing Python based data ingestion workflows to extract source-system data and organize raw files in Google Cloud Storage for downstream processing.
- Supported BigQuery data loading and table creation, preparing structured datasets, defining schemas, and validating records for analytical reporting.
- Developed basic Python data transformation scripts to handle CSV and JSON files, perform data cleansing, standardize column formats, and prepare datasets for warehouse ingestion.
- Assisted with SQL query development and validation in BigQuery, performing joins, aggregations, filtering, and data reconciliation to support business reporting requirements.
- Supported batch pipeline testing and troubleshooting, reviewing ingestion logs, identifying data inconsistencies, validating transformation outputs, and documenting issues for senior data engineers.

Graduate Teaching Assistant – Professor Jia Ronnie - Python Programming ***Aug 2023 – May 2024***
University of North Texas | Denton,TX

- Served as a Python instructor for undergraduate students, teaching core programming concepts including variables, data types, conditional statements, loops, functions, exception handling, file operations, and object-oriented programming.

- Assisted undergraduate students in learning Python programming fundamentals, including variables, data types, loops, conditional statements, functions, and basic data structures.
- Conducted hands-on programming sessions covering lists, tuples, dictionaries, string operations, file handling, and basic object oriented programming.

Graduate Teaching Assistant – Professor Vipul Garg - Supply Chain Management ***May 2023 – Aug 2023***
University of North Texas | Denton,TX

- Conducted human factors and occupational safety research involving forklift operators, collecting real time EEG brainwave signals using a multichannel EEG headset and first person video recordings through camera equipped smart glasses to study cognitive workload, attention, and operator responses.

- Developed data ingestion and preprocessing workflows to organize multichannel EEG time series data and video recordings, applying data cleaning, and structured transformations to prepare synchronized datasets for downstream analysis.

- Supported data processing pipelines for EEG research datasets, organizing recording sessions, validating collected data, and preparing transformed datasets for analysis of operator behavior, cognitive responses, and workplace safety conditions.

CERTIFICATIONS:

- AWS Certified Data Engineer – Associate

- Azure Databricks Data Engineer-Associate